CYBORG SUPERMARKET ENTERPRISE DOCUMENTATION

API Reference and Integration Guide

RESTful Contracts, OpenAPI 3.0 Specifications, and DTO Schemas

DOCUMENT ID: 05_API_Reference_and_Integration_Guide.pdf

Executive Publication Record

Organization:	Cyborg Omnichannel Engineering Organization
Ecosystem:	Windows POS • Android Mobile • Web Storefront • Admin Web • Backend
Primary Authority:	cyborg_system_architecture.pdf (16-Page Blueprint)
Source Dataset:	super_market(1).xlsx (1,000 Verified Egyptian Retail Records)
Security Level:	Enterprise Retail Distribution / Stakeholder Review
System Stack:	NestJS • TypeScript • PostgreSQL 16 • Redis 7 • SQLite • Flutter
Target Currency:	Egyptian Pounds (EGP) • Cairo Timezone (UTC+2 / UTC+3)

1. API Architecture & Standards

The Cyborg Supermarket API follows RESTful conventions under the versioned prefix /api/v1. Standard HTTP response envelopes encapsulate data payloads and pagination metadata:

- Authentication: Bearer JWT tokens via HTTP Authorization header. Tokens expire in 15 minutes.

- Rate Limiting: Enforced at 100 requests / minute per client IP using Redis token bucket.

- Idempotency: Critical write endpoints (e.g. POST /orders) accept an Idempotency-Key header.

2. Core Endpoints Inventory

Method	Endpoint Route	Required Role	Purpose
POST	/api/v1/auth/login	Public	Authenticate credentials and i
POST	/api/v1/auth/register	Public	Register customer account
GET	/api/v1/products	Public	Browse catalog with pagination
GET	/api/v1/products/{id}	Public	Get product details with real-
GET	/api/v1/products/barcode/{code	Public	Instant barcode lookup for POS
POST	/api/v1/orders	Authenticated	Submit new order (recalculated
PUT	/api/v1/orders/{id}/status	Manager/Cashier	Transition order status in sta
POST	/api/v1/shifts/start	Cashier/Manager	Open new cashier shift with op
PUT	/api/v1/shifts/{id}/close	Cashier/Manager	Close shift and calculate draw
POST	/api/v1/sync/ingest	Cashier/POS	Upload batch of offline POS tr
POST	/api/v1/ai/voice	Public	Parse colloquial Egyptian Arab